

ET AI HACKATHON 2.0

IndustrAI

Unified Asset & Operations Brain - Detailed Solution Document

Problem Statement 8 · IndustrAI

A working prototype, not a concept - every screenshot in this document is from the live application.

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Date: 22/07/2026
Place: Bengaluru Urban

1. Executive Summary

IndustrAI is a unified Industrial Knowledge Intelligence platform that ingests heterogeneous plant documentation - engineering drawings, maintenance records, safety procedures, inspection reports, operating instructions, and project files - and turns it into a single, continuously updated, queryable knowledge base.

The platform answers operational and engineering questions in plain language, grounds every answer in a cited source document, links related entities across the organisation's full document estate through a knowledge graph, and proactively flags contradictions between procedures before they cause a field mistake.

The platform is designed to work identically for a plant engineer at a desktop and a field technician on a mobile device, closing the knowledge gap that currently costs asset-intensive industries a significant share of their working hours and contributes materially to unplanned downtime.

2. Problem Context

Knowledge fragmentation in industrial operations is not a filing problem — it is a safety, quality, and operational efficiency problem that compounds over time.

35%

of working hours in asset-intensive industries are spent searching for information or recreating documents that already exist (McKinsey Global Survey, 2024)

7–12

disconnected document systems in the average large Indian plant P&IDs, work orders, procedures, inspection records, and regulatory filings, all siloed (NASSCOM–EY study)

18–22%

of unplanned downtime events in Indian heavy industry are attributable to this fragmentation (BIS Research)

25%

of India's experienced industrial engineers and operators will retire within the next decade, taking undocumented operational knowledge with them

Organisations that solve knowledge fragmentation first will have a structural advantage in how they operate, maintain, and improve their assets.

3. Challenge Statement

Build an AI-powered Industrial Knowledge Intelligence platform that ingests heterogeneous documents engineering drawings, maintenance records, safety procedures, inspection reports, operating instructions, project files across structured and unstructured formats, and makes their collective intelligence queryable, actionable, and continuously updated at the point of need, across any device or function.

4. Proposed Solution

IndustrAI addresses this challenge through four integrated capabilities built on a common retrieval and knowledge-graph layer:

4.1 Expert Knowledge Copilot

A RAG-powered conversational interface that answers operational, maintenance, and engineering queries across the full document corpus. Every response includes source citations, a confidence score, and a direct link to the originating document. The interface is built mobile-first so field technicians get the same capability as engineers at a desktop.

4.2 Knowledge Graph

An AI pipeline processes PDFs, P&IDs, scanned forms, spreadsheets, and email archives, extracting entities equipment tags, process parameters, regulatory references, personnel, dates and building a unified knowledge graph that maintains relationships across document types. The graph updates automatically as new records arrive, so the picture of any asset stays current without manual re-indexing.

4.3 Contradiction Detection

The system continuously compares guidance across documents describing the same equipment or procedure and flags conflicts for example, a safety procedure that has not been updated to match a revised OEM manual. This surfaces the kind of inconsistency that today only becomes visible after an incident.

4.4 Document-Type Filtering

Users can scope any search to a specific document class manuals, regulatory filings, inspection forms, or P&ID drawings, so a query returns answers from the right authority level rather than mixing informal notes with certified procedures.

5. System Architecture

The platform is split into an offline ingestion pipeline and an online query pipeline, both backed by a shared vector store and knowledge graph.

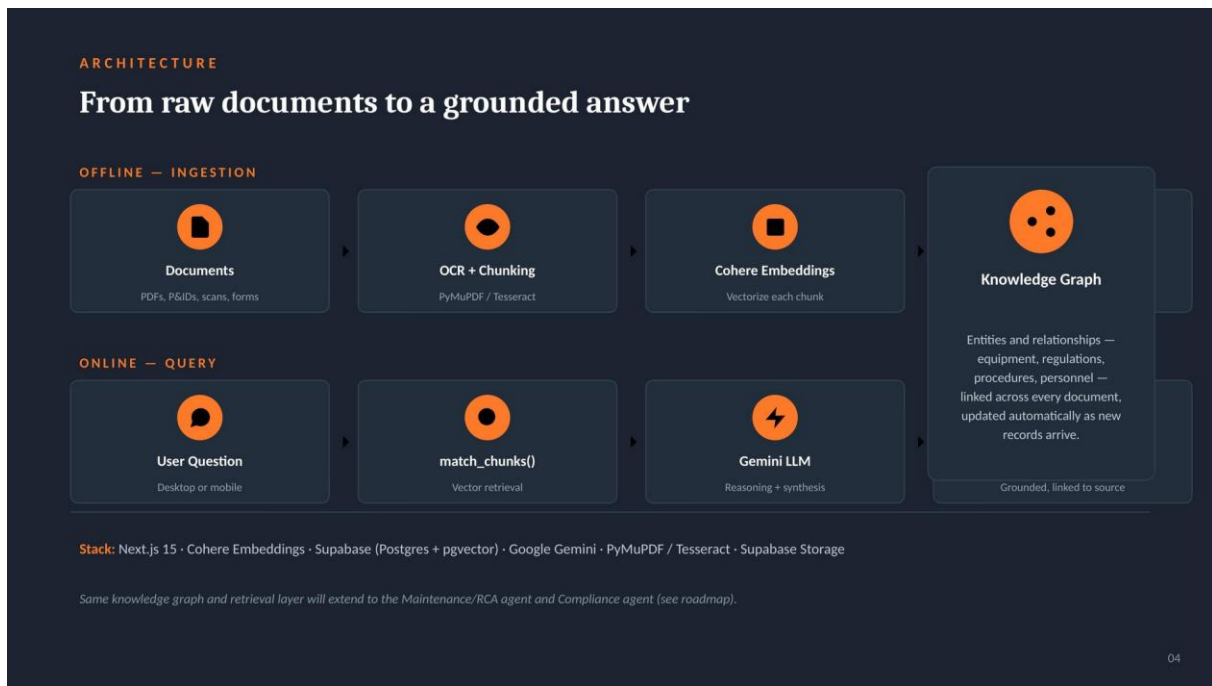


Figure 1. IndustrAI system architecture offline ingestion and online query pipelines sharing a common Supabase data layer.

5.1 Offline - Ingestion

- Documents (PDFs, P&IDs, scanned forms) enter the pipeline from Supabase Storage.
- OCR and chunking (PyMuPDF / Tesseract) extract text and structure from scanned and native documents.
- Cohere embeddings vectorize each chunk for semantic retrieval.
- Vectors are indexed in Supabase Postgres with the pgvector extension.
- Entities and relationships extracted during ingestion are written into the knowledge graph, linking equipment, regulations, procedures, and personnel across every document.

5.2 Online - Query

- A user submits a natural-language question from desktop or mobile.
- A `match_chunks()` vector retrieval call pulls the most relevant chunks from pgvector.
- Google Gemini performs reasoning and synthesis over the retrieved context.
- The response is returned with citations and links back to the originating source document.

6. Live Prototype

IndustrAI is a fully functioning application today, not a mockup. The screenshots below are captured directly from the running platform.

6.1 Expert Knowledge Copilot (Chat)

The core conversational interface lets users ask natural-language questions across the full document corpus and receive grounded, cited answers in real time. Every response includes the source document, page number, and similarity score for each citation, so answers are fully traceable back to the original manual, standard, or inspection record never a black-box answer.

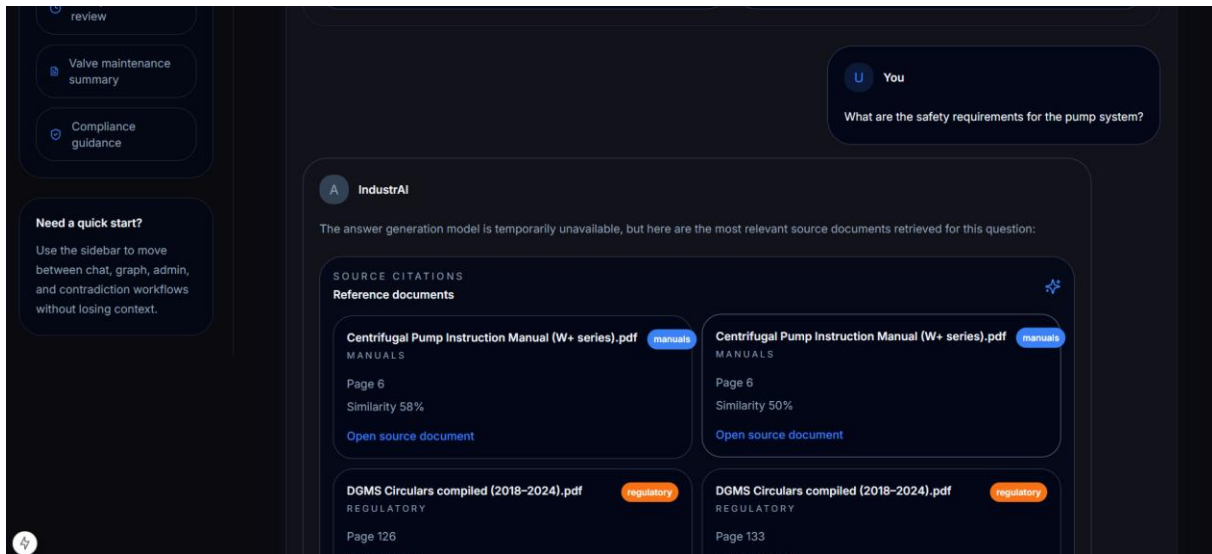


Figure 2. Expert Knowledge Copilot answering a safety query with cited, page-linked sources from the ingested document corpus.

6.2 Document Ingestion & Admin

Documents are uploaded, classified by type, chunked, and indexed with per-document status visible to the operator.

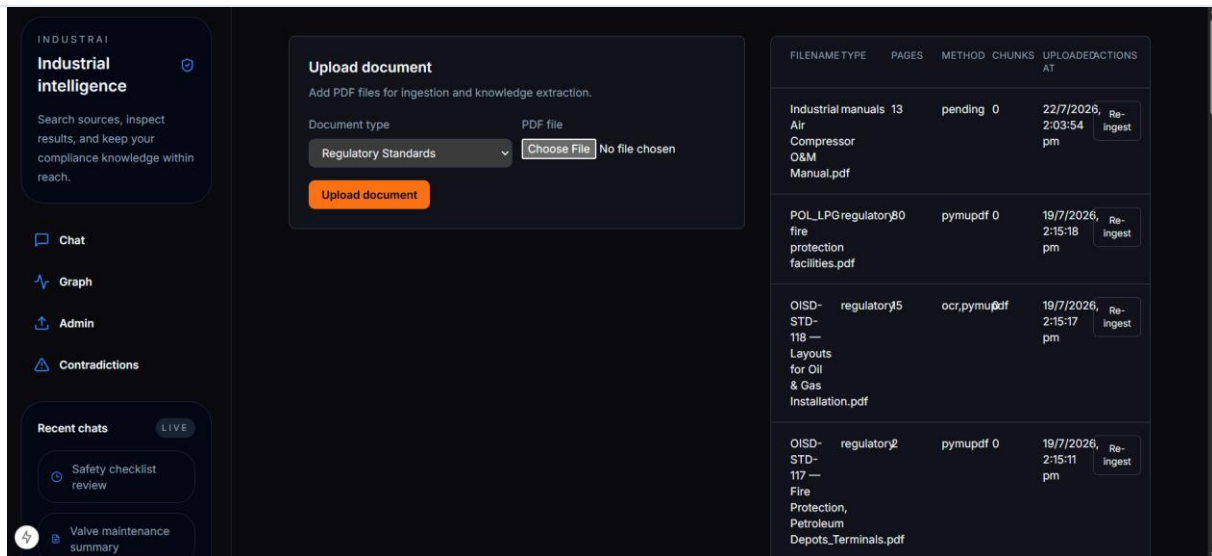


Figure 3. Admin panel showing ingested documents with type, page count, extraction method, and chunk counts.

6.3 Knowledge Graph

Equipment, regulation clauses, procedures, and locations extracted from the corpus are automatically linked into an explorable graph.



Figure 4. Knowledge graph view equipment, regulation clauses, and procedures linked across ingested pump manuals.

6.4 Contradiction Detection

The contradiction detector runs on demand and reports its findings directly in the UI, with severity filtering (high / medium / low).

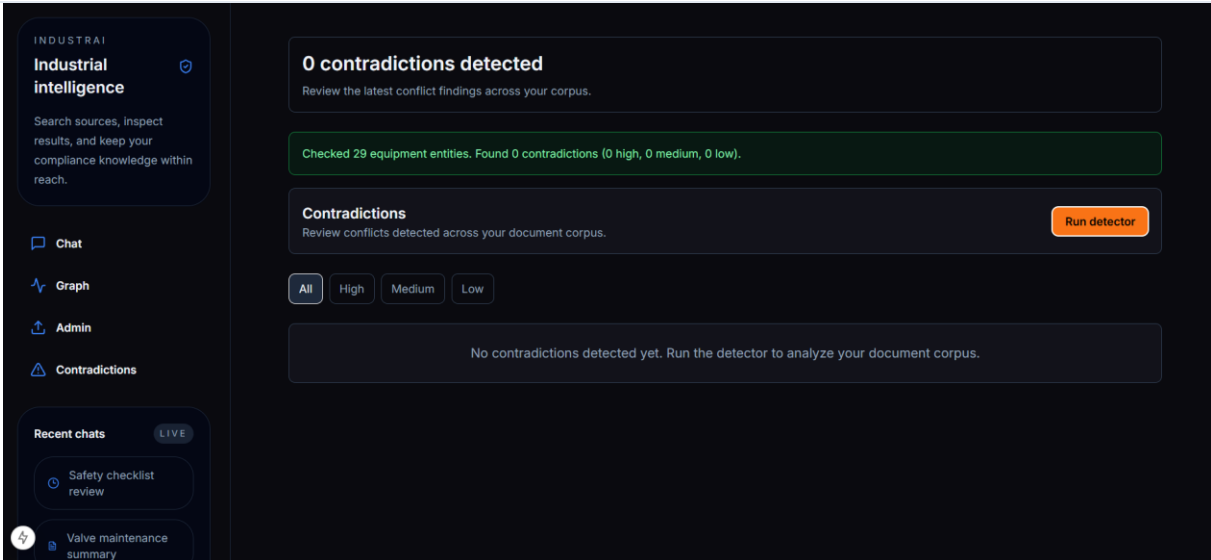


Figure 5. Contradiction detection panel with live run status and severity filters.

7. Technology Stack

Layer	Technology	Purpose
Application framework	Next.js 15	Frontend and API routes
Embeddings	Cohere	Vectorizing document chunks
Database & vector store	Supabase (Postgres + pgvector)	Structured data and semantic search
LLM reasoning	Google Gemini	Answer synthesis and grounding
OCR & extraction	PyMuPDF / Tesseract	Text extraction from scans and native PDFs
Document storage	Supabase Storage	Raw document repository

8. Judging Criteria Alignment

Criterion	Weight	How IndustrAI addresses it
Innovation	25%	Cross-document knowledge graph combined with contradiction detection — not just single-document RAG.
Business Impact	25%	Directly targets the 35% search-time waste and 18–22% downtime attributable to knowledge fragmentation.
Technical Excellence	20%	Full RAG pipeline with OCR fallback, vector search, grounded generation, and entity extraction — running end to end in a live prototype.
Scalability	15%	Cloud-native, API-based LLMs, horizontally scalable ingestion pipeline.

UX	15%	Conversational interface, voice input, mobile-friendly design, and cited sources on every answer.
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9. Evaluation Approach

In line with the stated evaluation focus for this problem statement, the prototype is designed to be assessed on:

- Entity extraction accuracy across document types
- Query answer quality against domain-expert benchmark questions
- Knowledge graph linkage completeness
- Time-to-answer compared to traditional keyword search
- Compliance gap detection accuracy
- Demonstrated improvement in cross-functional knowledge discovery, validated against real industrial document samples

10. Roadmap

The current prototype establishes the retrieval and knowledge-graph foundation. Planned extensions build directly on that same layer:

- Expand entity extraction across the full document corpus, beyond the initial sample set
- Add computer vision for automated P&ID symbol parsing and drawing digitisation
- Build a Maintenance Intelligence & RCA agent that fuses work-order history, failure records, and OEM manuals for predictive maintenance recommendations
- Build a Compliance Intelligence module that maps OISD, PESO, Factory Act, and environmental norms against current procedures and equipment states, auto-generating audit evidence packages

11. Conclusion

IndustrAI turns a plant's fragmented document estate into a single, continuously updated operational brain cutting the time engineers and technicians spend searching for information, surfacing contradictions before they cause incidents, and preserving institutional knowledge as experienced staff retire.

The architecture is deliberately built on a shared retrieval and knowledge-graph layer so that the roadmap items maintenance intelligence, compliance mapping, and computer-vision-based P&ID parsing are extensions of the same core system rather than separate products.